

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – IX
PAPER –1
TEST DATE: 17-08-2022

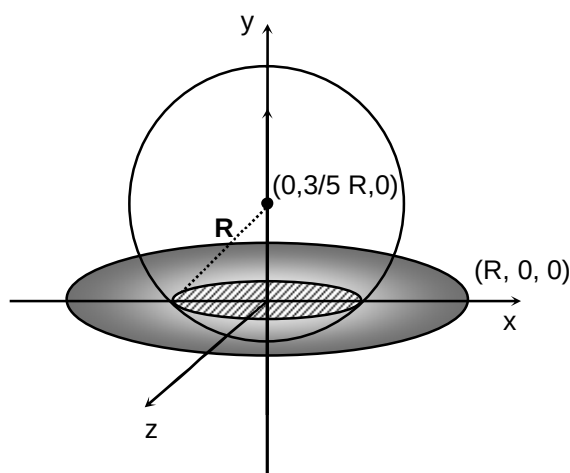
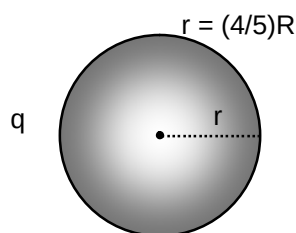
ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B
 Sol.



Charge on circular part

$$q = \int \sigma(r) \cdot 2\pi r \cdot dr$$

$$= \int_0^{(4/5)R} 2\pi\sigma_0 \left(r - \frac{r^3}{R^2} \right) dr = \frac{272}{625} (\sigma_0 \pi R^2)$$

$$\phi = \frac{q}{\epsilon_0} = \frac{272}{625} \left(\frac{\sigma_0 \pi R^2}{\epsilon_0} \right)$$

2. C
 Sol.

The path of projectile motion as observed by the other projectile motion is a straight line. The vertical component 30 m/s will get cancelled. Hence B will travel horizontal with respect to A. Hence minimum separation = 20 m.

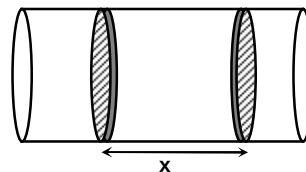
3. A

 Sol. $P_1 V_1 = P_2 V_2$

$$(10^5) (20,000 \times 10^{-6}) = \left\{ 10^5 + \frac{(2k \cos^2 37^\circ)x}{100 \times 10^{-4}} \right\} (100 \times 10^{-4})x$$

$$x^2 + x - 2 = 0$$

$$x = 1 \text{ m} = 100 \text{ cm}$$



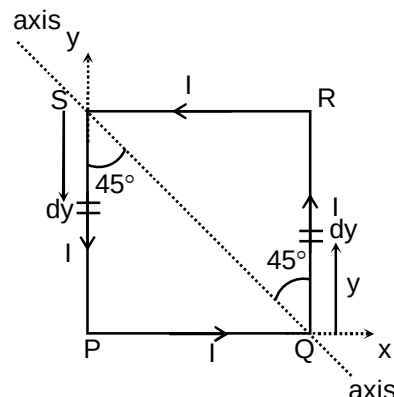
4. A

Sol. Torque on the loop due to magnetic field

$$\tau = \int_0^\ell (I B_0 dy) \frac{y}{\sqrt{2}} + \int_0^\ell (I B_0 dy) \frac{y}{\sqrt{2}}$$

$$I\alpha = \frac{3IB_0\ell^2}{2\sqrt{2}}$$

$$\frac{1}{6} m \ell^2 \alpha = \frac{3IB_0\ell^2}{2\sqrt{2}} \Rightarrow \alpha = \left(\frac{9IB_0}{\sqrt{2}m} \right)$$



5. A, B, D

$$\text{Sol. } f = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2} = \frac{3 \times 5 + 2 \times 3}{3 + 2} = \frac{21}{5}$$

$$\gamma = 1 + \frac{2}{f} = \frac{31}{21}$$

 Let initial volume and pressure are V_0 and P_0 respectively

Heat absorb in isobaric expansion

$$Q_1 = \frac{\gamma}{\gamma - 1} (3P_0 V_0) = 9.3P_0 V_0$$

Heat rejected in isochoric cooling

$$Q_2 = \frac{1}{\gamma - 1} (3P_0 V_0) = 6.3P_0 V_0$$

 Heat rejected in isothermal compression $Q_3 = nRT_0 \ln 4 = P_0 V_0 \ln 4$

$$\text{Efficiency} = \frac{Q_1 - Q_2 - Q_3}{Q_1} = \left(\frac{161}{930} \right)$$

6. A, D

$$\text{Sol. } \frac{1}{2} \mu u^2 - \frac{ke^2}{r} = 0$$

$$\Rightarrow u = \sqrt{\frac{2k}{\mu r}} e$$

$$\lambda = \frac{h}{\mu u} \propto \frac{1}{u} \propto \sqrt{r}$$

$$KE_1 = \frac{1}{2} \mu u^2 = \frac{ke^2}{r}$$

$$\text{For Bohr model, } \frac{1}{2} m v^2 = \frac{ke^2}{2r} = KE_2$$

$$\Rightarrow \mu = \frac{KE_1}{KE_2} = 2$$

7. A, B, C

Sol. The deviation produced by ABCD is zero. Hence the cavity will not have any effect on the deviation.

$$\mu = \frac{\sin\left(\frac{\delta_{\min} + A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{8}{5}$$

$$\sin\left(\frac{\delta_{\min} + A}{2}\right) = \left(\frac{8}{5}\right) \sin\left(\frac{60^\circ}{2}\right)$$

$$\delta_{\min} = 46^\circ$$

8. A, C

Sol. Equivalent resistance between A and E

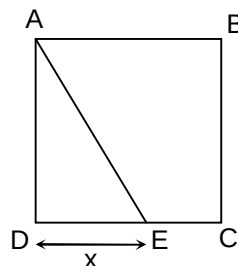
$$y = \frac{x + 22}{x + 22 + 32}$$

For B and E to be equipotential

$$\frac{R_{AE}}{R_{AR}} = \frac{R_{EC}}{R_{RC}}$$

$$\Rightarrow \frac{(22 + x) \times 32}{(54 + x) \times (14 - x)} = \frac{8}{2} = 4$$

Solve to get : $x = 10 \Omega$



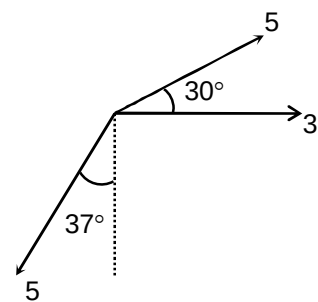
9. A, C, D

$$\text{Sol. } z = \sqrt{(2-1)^2 + \left(\frac{2}{\sqrt{3}}\right)^2} = \sqrt{\frac{7}{3}} \Omega$$

$$V_1 = 3 \sin \omega t, V_2 = 5 \sin(\omega t + 30^\circ), V_3 = 5 \sin(\omega t - 127^\circ)$$

$$V_{\max} = \sqrt{21} \text{ volts}$$

$$I_{\max} = \frac{V_{\max}}{z} = \frac{\sqrt{21}}{\sqrt{\frac{7}{3}}} = 3 \text{ amp}$$



10. B, C, D

$$\text{Sol. For A, } 1V = \frac{7S}{10} = 0.7 \text{ mm}$$

$$\text{Least count of A} = 5 - 0.7 \times 7 = 0.1 \text{ mm}$$

$$\text{For B, } 1V = \frac{3S}{8} = 0.375 \text{ mm}$$

$$\text{Least count of B} = 2 - 0.375 \times 5 = 0.125$$

$$\text{Difference} = 0.125 - 0.1 = 0.025 \text{ mm}$$

Section – B

11. 4

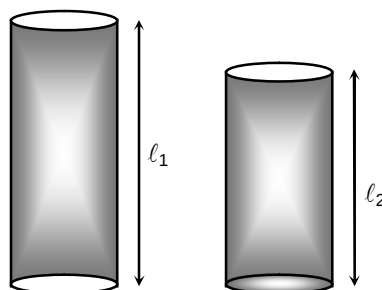
Sol. $R = \frac{28 \times 42}{70} = 16.8 \Omega$

12. 4

Sol. $v = 4t - 3t^2$
 $S = 2t^2 - t^3 + C$
 When $v = 0, \Rightarrow t = 0$ and $t = 4/3$
 $S(t = 0) = C, S\left(t = \frac{4}{3}\right) = \frac{32}{27} + C, S(t = 2) = 0$
 Distance traveled $= \frac{32}{27} + \frac{32}{27} = \frac{64}{27}$
 Average speed $= \frac{64}{27 \times 2} = \frac{32}{27} \text{ m/s}$

13. 2

Sol. $\frac{3v}{2\ell_1} = \frac{5v}{4\ell_2}$
 $\frac{3v}{2(\ell_1 + 2e)} = \frac{5v}{4(\ell_2 + e)}$
 $e = \frac{12\ell_2 - 10\ell_1}{8} = 2 \text{ cm}$



Section – C

14. 288.00

15. 32.00

Sol. (for Q.14-15):

 When S_1, S_2 both are open capacity of the system is C_1 . Then C_1 .

$$C_1 = \frac{\frac{4\pi\epsilon_0(3R)(R)}{(3R-R)}(4\pi\epsilon_0 4R)}{\frac{4\pi\epsilon_0(3R)(R)}{(3R-R)} + 4\pi\epsilon_0 4R}$$

$$C_1 = \frac{(4\pi\epsilon_0 R)\left(\frac{3}{2}\right)(4)}{\left(\frac{3}{2} + 4\right)} = \frac{(4\pi\epsilon_0 R)(3.4)}{11}$$

$$C_1 = \frac{48\pi\epsilon_0 R}{11}$$

 When S_1 is closed but S_2 is open capacity is C_2 . Find C_2 .

$$C_2 = \frac{\frac{4\pi\epsilon_0(3R)(2R)}{(3R-2R)}(4\pi\epsilon_0 \cdot 4R)}{\frac{4\pi\epsilon_0(3R)(2R)}{(3R-2R)} + 4\pi\epsilon_0 4R} = \frac{(4\pi\epsilon_0 R)(6)(4)}{(6+4)} = \frac{48\pi\epsilon_0 R}{5}$$

When S_2 is closed but S_1 is opened capacity is C_3 . Find C_3

$$C_3 = \frac{\frac{4\pi\epsilon_0(2R)(R)}{(2R-R)}(4\pi\epsilon_0 4R)}{\frac{4\pi\epsilon_0(2R)(R)}{(2R-R)} + 4\pi\epsilon_0 4R} = \frac{4\pi\epsilon_0 R \cdot 2 \cdot 4}{6} = \frac{16\pi\epsilon_0 R}{3}$$

When both S_1 and S_2 are closed. Capacity is C_4 is . Then C_4 is equal to

$$C_4 = 4 \pi \epsilon_0 4R$$

$$C_4 = 16 \pi \epsilon_0 R$$

16. 25.00

17. 25.00

Sol. **(for Q.16-17):**

When B and D is cut

$$mg - T = ma$$

taking torque about mid point of rod

$$T \cdot \frac{\ell}{2} = \frac{m\ell^2}{12} \times \frac{a}{\ell/2}$$

$$T = mg/4$$

18. 02.00

19. 03.00

Sol. **(for Q.18-19):**

There is a dark fringe at O if path difference = $\lambda/2$

$$\Delta x = SO_2O - SO_1O = \frac{\lambda}{2}$$

$$2\sqrt{D^2 + d^2} - 2D = \frac{\lambda}{2}$$

$$2D\left(1 + \frac{d^2}{2D^2}\right) - 2D = \left(\frac{\lambda}{2}\right)$$

$$d_{\min} = \sqrt{\frac{D\lambda}{2}}$$

The bright fringe is formed at P if the path difference = λ

$$\Delta x = SO_1P - SO_2P = \lambda$$

$$\left\{ \left(D + \sqrt{D^2 + x^2} \right) - \left(\sqrt{D^2 + d^2} + \sqrt{D^2 + (x-d)^2} \right) \right\} = \lambda$$

$$D + D\left(1 + \frac{x^2}{2D^2}\right) - D\left(1 + \frac{d^2}{2D^2}\right) - D\left(1 + \frac{(x-d)^2}{2D^2}\right) = \lambda$$

$$\frac{x^2}{2D} - \frac{d^2}{2D} - \frac{(x-d)^2}{2D} = \lambda$$

$$x = 2\sqrt{\frac{\lambda D}{2}} = 3 d_{\min}$$

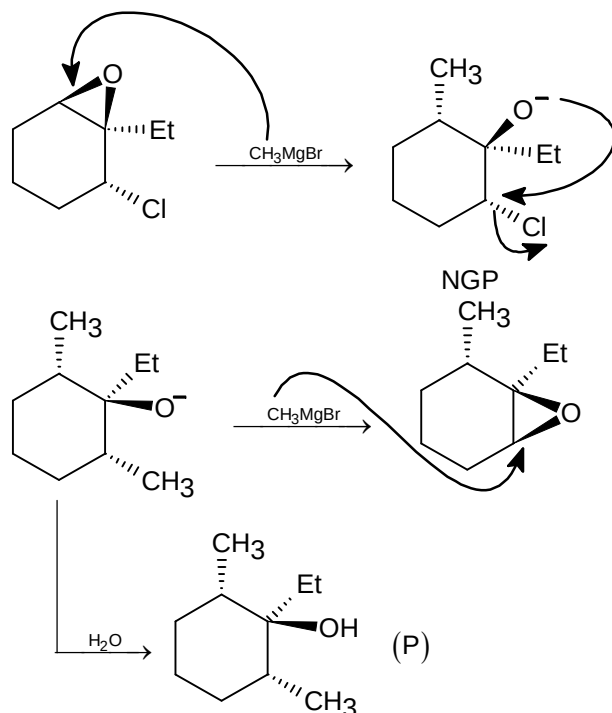
Chemistry

PART – II

Section – A

20. D
Sol. Due to absence of plane of symmetry. Option (D) is optically active.

21. A
Sol.



22. A
Sol. $\text{KI} + \text{O}_3 \longrightarrow \text{KOH} + \text{I}_2 + \text{O}_2$
 $\text{I}_2 + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{NaI} + \text{Na}_2\text{S}_4\text{O}_6$
 Law of chemical equivalence,
 $(\text{ne})_{\text{I}_2} = (\text{ne})_{\text{Na}_2\text{S}_2\text{O}_3} = (\text{ne})_{\text{O}_3}$
 $\therefore (\text{ne})_{\text{Na}_2\text{S}_2\text{O}_3} = (\text{ne})_{\text{O}_3}$
 $\Rightarrow 2 \times 1 = 2 \times [\text{mole}]_{\text{O}_3}$
 $[\text{mole}]_{\text{O}_3} = 1$
 $\text{Mole of } \% \text{ O}_3 = \frac{1}{4} \times 100 = 25\%$

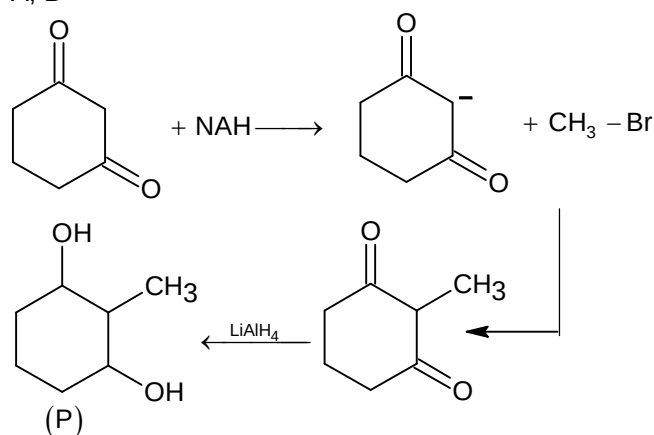
23. C
Sol. $\text{XeF}_2 \longrightarrow \text{linear}$
 $\text{ICl}_4^- \longrightarrow \text{square planar}$
 $\text{XeF}_5^+ \longrightarrow \text{square pyramidal}$
 $\text{XeF}_5^- \longrightarrow \text{pentagonal planar}$

24. A, B

Sol. A and B are high spin complex [NO pairing] C and D are low spin complex.

25. A, B

Sol.



Number of stereo isomers in P is 4.

26. A, C, D

Sol. $\psi_{n,\ell,m} \propto \left(\frac{Z}{a}\right)^{3/2} \left(\frac{Zr}{a}\right) e^{-2r/2a} \cos(\theta)$ and xy plane is nodal plane

Number of nodal plane = 1

 $2p_z$ is ungerade orbital.

27. A, B, D

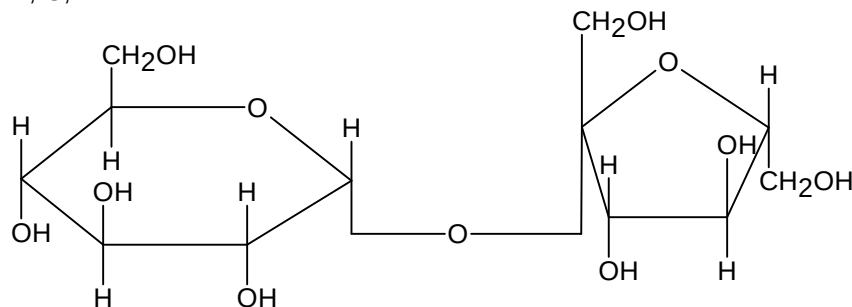
Sol. $[\text{Ag}(\text{NH}_3)_2][\text{AgCl}_2]$ will not show the co-ordination isomerism.

28. B, C

Sol. Polythene and polypropene are homopolymer.

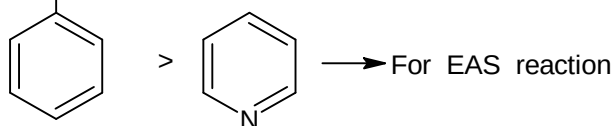
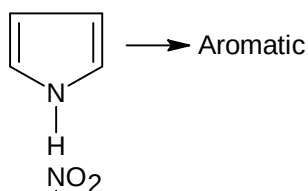
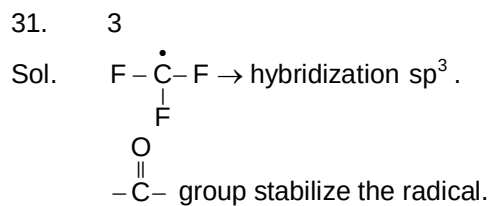
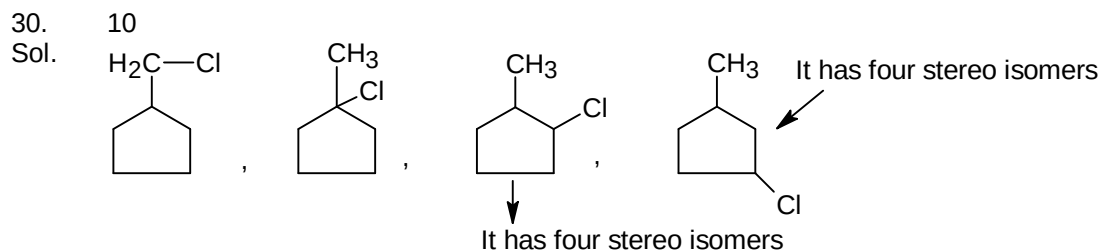
29. B, C, D

Sol.

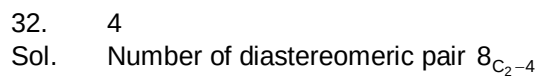


Structure is of sucrose which is non reducing sugar due to absence of Hemiacetal group.

Section – B



$[\Delta S]_{\text{system}} > 0$ for adiabatic irreversible process.



$$= \frac{8}{2 \times 6} - 4$$

$$= \frac{7 \times 8}{2} - 4$$

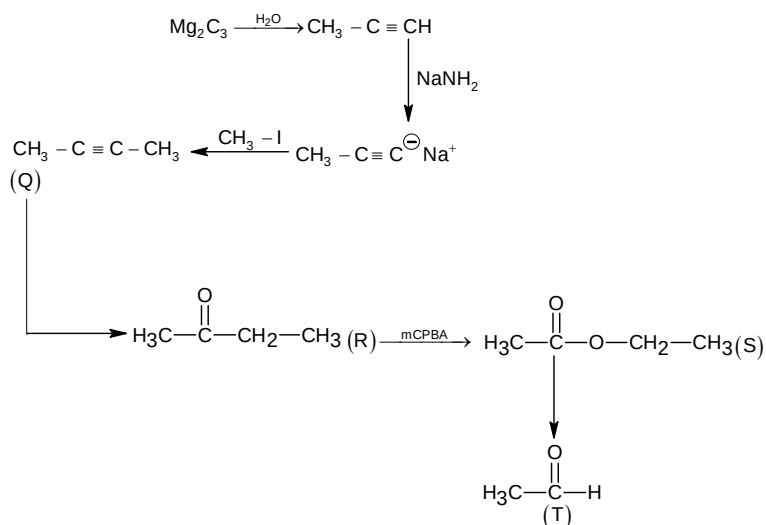
$$x = 24$$

$$\therefore \frac{x}{6} = 4$$

Section – C

33. 10.08

34. 03.44
Sol. (for Q. 33 to 34):



$$\therefore [\text{Mass}]_{\text{CH}_3 - \text{C} \equiv \text{C} - \text{H}} = 10, [\text{M}]_{\text{CH}_3 - \text{C} \equiv \text{C} - \text{H}} = 40 \text{ gram/mole}$$

$$\therefore [\text{Mole}]_{\text{CH}_3 - \text{C} \equiv \text{CH}} = \frac{10}{40} = \frac{1}{4}$$

$$\therefore \text{Mole of Q} = \frac{1}{4} \times \frac{70}{100} = \frac{7}{40}$$

$$\therefore \text{Mole of R} = \frac{7}{40} \times \frac{80}{100}$$

$$\therefore \text{Mass of R} = \frac{7}{40} \times \frac{80}{100} \times [72]$$

$$x = 10.08 \text{ g}$$

$$\text{Mass of T} = \frac{7}{40} \times \frac{80}{100} \times \frac{80}{100} \times \frac{70}{100} \times [44]$$

35. 01.00

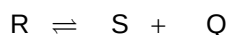
36. 04.00

Sol. (for Q. 35 to 36):



$$2 \quad 0 \quad 0$$

$$2 - x \quad x + y \quad x - y$$



$$x \quad 0 \quad 0$$

$$x - y \quad y \quad y + x$$

$$2 - x + x + y + x - y + y = 4$$

$$x + y = 2$$

$$\frac{x - y}{x + y} = \frac{1}{5}; y = \frac{2}{3}x$$

$$\text{Hence, } x = 1.2$$

$$y = 0.8$$

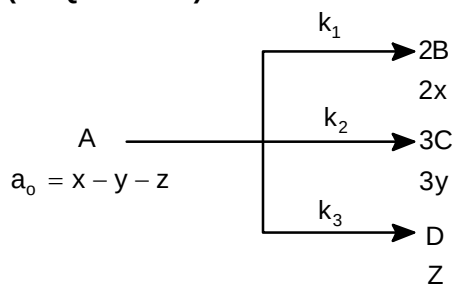
$$K_{C_1} = W = \frac{(x - y)(x + y)}{2 - x} = \frac{2 \times 0.4}{0.8} = 1$$

$$K_{C_2} = Z = \frac{(x+y)(y)}{x-y} = \frac{2 \times 0.8}{0.4} = 4$$

37. 01.08

38. 00.36

Sol. (for Q. 37 to 38):



$$A_t = a_0 \times e^{-(k_1+k_2+k_3)t}$$

$$= 1 \times e^{-[0.5]4.606}$$

$$\therefore A_t = [10]^{-1} = \frac{1}{10} = 0.1 \text{ M}$$

$$\frac{k_1}{k_2} = \frac{x}{y} \Rightarrow \frac{1}{2} = \frac{x}{y} \quad \dots (1)$$

$$\frac{k_2}{k_3} = \frac{y}{z} \Rightarrow y = z \quad \dots (2)$$

$$A_t = a_0 - [x + y + z]$$

$$0.1 = 1 - [x + y + z]$$

$$x + y + z = 1 - 0.1 = 0.9 \quad \dots (3)$$

$$A_t = 0.1 \text{ M}, B_t = 2x = 0.36 \text{ M}$$

$$C_t = 3y = 1.08 \text{ M}, D_t = 0.36 \text{ M}$$

Mathematics**PART – III****Section – A**

39. C

Sol. Let $(x^2 + 2x)^{\frac{1}{3}} = y \Rightarrow x^2 + 2x + 1 = y^3 + 1 \Rightarrow x = -1 + (y^3 + 1)^{\frac{1}{3}} \Rightarrow f^{-1}(x) = -1 + (x^3 + 1)^{\frac{1}{3}}$

$$\int_0^2 (f^{-1}(x) + 1 + f(x)) dx \Rightarrow \int_0^2 f^{-1}(x) dx$$

Let $f^{-1}(x) = t \Rightarrow x = f(t) \Rightarrow \frac{dx}{dt} = f'(t) ; \int tf'(t) dt = tf(t) - \int f(t) dt = 2f(2) - \int_0^2 f(x) dx$

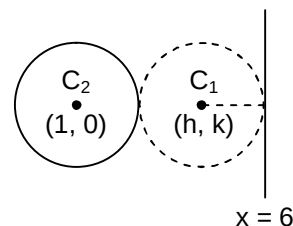
$$\Rightarrow \int_0^2 (f^{-1}(x) + 1 + f(x)) dx = 2f(2) - \int_0^2 f(x) dx + \int_0^2 1 dx + \int_0^2 f(x) dx = 2f(2) + 2 = 2 \times 2 + 2 = 6$$

40. B

Sol. $10m^2 + 5 - c^2 < 0$
Thus $c^2 > 15$

41. D

Sol. $C_1 C_2 = r_1 + r_2 \Rightarrow \sqrt{(h-1)^2 + k^2} = r + 3$ and $h = 6 - r$
 $\Rightarrow r = 6 - h ; (h-1)^2 + k^2 + r^2 + 9 + 6r$
 $\Rightarrow h^2 + 1 - 2h + k^2 = (6-h)^2 + 9 + 6(6-h)$
 $\Rightarrow h^2 + 1 - 2h + k^2 = 36 + h^2 - 12h + 9 + 36 - 6h$
 $\Rightarrow k^2 = 16h + 80 \Rightarrow y^2 = -16(x-5)$
Equation of directrix is $x - 5 = 4 \Rightarrow x = 9$



42. B

Sol. Let $g(x) = f(x)(x+1) - x \Rightarrow g(0) = g(1) = g(2) = g(3) = g(4) = 0$
 $\therefore g(x) = f(x)(x+1) - x = kx(x-1)(x-2)(x-3)(x-4)$
Put $x = -1 \Rightarrow 1 = kx - 1x - 2x - 3x - 4x - 5 \Rightarrow k = -\frac{1}{120}$
 $\therefore f(x)(x+1) - x = -\frac{x(x-1)(x-2)(x-3)(x-4)}{120}$
Put $x = 6 \Rightarrow f(6) \times (7-6) = -\frac{6 \times 5 \times 4 \times 3 \times 2}{120} \Rightarrow f(6) = 0$

43. B, D

Sol. $\int_{-\infty}^k \frac{(\sin^{-1} e^x + \cos^{-1} e^x) e^x}{(\tan^{-1} e^k + \tan^{-1} e^x) e^{2x} + 1} dx$. Let $\tan^{-1} e^x = t$
 $\frac{\pi}{2} \int_0^{\tan^{-1}(e^k)} \frac{1}{(t + \tan^{-1} e^k)} dt$

44. A, B

Sol. $\alpha^2 = x \Rightarrow \alpha = \sqrt{x}$ replace x by $\sqrt{x}$ in $x^3 + x + 1 = 0$. We get $g(x) = x^3 + 2x^2 + x - 1$
 $\Rightarrow g(3) = 47 \Rightarrow g(1) = 3 \Rightarrow 3^{50} = (10-1)^{25} = {}^{25}C_0 10^{25} + \dots + {}^{25}C_{23} 10^2 + {}^{25}C_{24} 10 - 1$
 $\Rightarrow g'(x) > 0 \forall x \in \mathbb{R} \therefore g(x)$ has only 1 real root

45. C, D

 Sol. $(PQP^T)^T = PQ^T P^T \Rightarrow PQP^T$ is not symmetric
 $(PQ - QP)^T = Q^T P^T - P^T Q^T$

$$Q = |P| \frac{\text{adj } P}{|P|} \Rightarrow Q = \text{adj } P$$

$$\Rightarrow (\text{adj } (P^T) - Q)^T = (\text{adj } (P^T) - \text{adj } P)^T = \text{adj } P - \text{adj } P^T$$

and $Q = -P^T \Rightarrow Q = P \Rightarrow Q$ is also skew symmetric

46. A, B, C, D

 Sol. Equation of L_1

$$5x - 4 \times y \times \frac{3}{2} = 16$$

$$5x - 6y = 16$$

 $\therefore$ Equation of tangent of B

$$5x - 6y = -16$$

 $\therefore$ Equation of normal at A

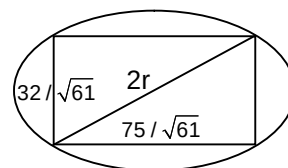
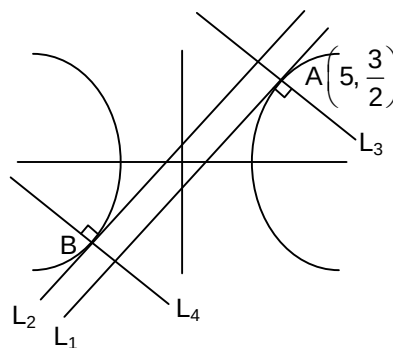
$$6x + 5y = \frac{75}{2}$$

 $\therefore$ Equation of normal at B is

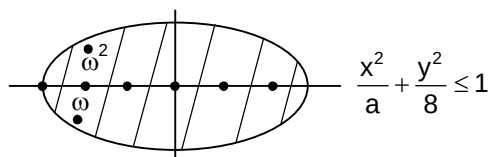
$$6x + 5y = -\frac{75}{2}$$

$$\text{Distance between tangents} = \frac{32}{\sqrt{61}}$$

$$\text{Distance between normals} = \frac{75}{\sqrt{61}}$$



47. A, C

 Sol. $||z + \omega^2| - |z + \omega||$ is minimum when z lies on the major axis of the ellipse


48. A, B, C

 Sol. $\int_1^x f(x) dx = \frac{f(x)}{2} \left(x + \frac{1}{x} \right) - \frac{x^2 + 1}{2} + 1$ using Leibnitz Rule

$$f(x) = \frac{f'(x)}{2} \left(x + \frac{1}{x} \right) + \frac{f(x)}{2} \left(1 - \frac{1}{x^2} \right) - x \Rightarrow \frac{xf'(x) - f(x)}{x^2} = \frac{2x}{x^2 + 1}$$

$$\text{Integrating both sides, we get } \frac{f(x)}{x} = \ln(x^2 + 1)$$

Section – B

49. 9

Sol. Let x_1 be the number of times he took 1 step up the staircase
 Let x_2 be the number of times he took 2 step up the staircase
 $x_1 + 2x_2 = 11$

			Number of ways
$x_1 = 1$	$x_2 = 6$	1, 2, 2, 2, 2, 2	$\frac{6!}{1! 5!} = 6$
$x_1 = 3$	$x_2 = 4$	1, 1, 1, 2, 2, 2, 2	$\frac{7!}{3! 4!} = 35$
$x_1 = 5$	$x_2 = 3$	1, 1, 1, 1, 1, 2, 2, 2	$\frac{8!}{3! 4!} = 56$
$x_1 = 7$	$x_2 = 2$	1, 1, 1, 1, 1, 1, 1, 2, 2	$\frac{9!}{2! 7!} = 36$
$x_1 = 9$	$x_2 = 1$		$\frac{10!}{9! 1!} = 10$
$x_1 = 11$	$x_2 = 0$		1
			144

$$\therefore P = 144$$

50. 4

Sol. $2 \sin A \sin B + 2 \sin B \sin C + 2 \sin C \sin A = \frac{9}{2}$

$$\cos(A - B) - \cos(A + B) + \cos(B - C) - \cos(B + C) + \cos(C - A) - \cos(C + A) = \frac{9}{2}$$

$$\Rightarrow \cos(A - B) + \cos(B - C) + \cos(C - A) + \cos A + \cos B + \cos C = \frac{9}{2}$$

$$\Rightarrow \cos(A + B) + \cos(B - C) + \cos(C - A) = \frac{9}{2} - (\cos A + \cos B + \cos C) \geq \frac{9}{2} - \frac{3}{2} \geq 3$$

$$\Rightarrow \cos(A - B) + \cos(B - C) + \cos(C - A) = 3$$

$$\Rightarrow A = B = C \therefore A = \frac{\sqrt{3}}{4} \times 16 = 4\sqrt{3}$$

51. 6

Sol. $(x - 2)(x^2 - 9x + 20) \geq 0 \Rightarrow x \in [2, 4] \cup [5, \infty)$

$$\therefore p = 2$$

$$\Rightarrow \left(\sin x + \frac{1}{2} \right)^2 + \frac{3}{4}, \text{ least value is } \frac{3}{4} \therefore q = \frac{3}{4}$$

$$\Rightarrow x = 1 \text{ satisfies it } \therefore \alpha = 1 \text{ and other root is } \frac{-1}{(2021)^{2022} - 1}$$

$$\therefore \alpha = 1, q = \frac{3}{4}, p = 2$$

Section – C

52. 2023.00

53. 1348.00

Sol. (for Q.52.-53):

$$A_1 = [k] \Rightarrow B_1 = k; A_2 = \begin{bmatrix} k & 1 \\ k & 1 \end{bmatrix} \Rightarrow B_2 = k^2 - 1; A_3 = \begin{bmatrix} k & 1 & 0 \\ 1 & k & 1 \\ 0 & 1 & k \end{bmatrix} \Rightarrow B_3 = KB_2 - 1B_1$$

$$\text{Similarly, } A_4 = \begin{bmatrix} k & 1 & 0 & 0 \\ 1 & k & 1 & 0 \\ 0 & 1 & k & 1 \\ 0 & 0 & 1 & k \end{bmatrix} \Rightarrow B_4 = kB_3 - B_2$$

 Thus, $B_n = KB_{n-1} - B_{n-2}$. If $k = 2$, $B_1 = 2$, $B_2 = 3$, $B_3 = 4$, $B_4 = 5$

 Thus, $B_{2022} = 2023$

 If $k = 1$, $B_n = B_{n-1} - B_{n-2}$; $B_1 = 1$, $B_2 = 0$, $B_3 = B_2 - B_1 = -1$, $B_4 = B_3 - B_2 = -1$
 $B_5 = B_4 - B_3 = 0$; $B_6 = 1$ and it goes on

 $B_{6n+1} = 1$; $B_{6n+2} = 0$; $B_{6n+3} = -1 \Rightarrow B_{6n+4} = -1$; $B_{6n+5} = 0$; $B_{6n} = 1$

$$\text{Thus, } \sum_{k=1}^{2022} |B_p| = \frac{4}{6} \times 2022 = 1348$$

54. 12.00

55. 24.00

Sol. (for Q. 54.-55):

 $z_1, z_2, z_3, \dots, z$ are also the roots of $z^{12} = 1$ i.e. 12th roots of unity

$$z^{12} - 1 = (z - z_1)(z - z_2)(z - z_3) \dots (z - z_{12})$$

$$\frac{z^{12} - 1}{z - 1} = (z - z_2)(z - z_3)(z - z_4) \dots (z - z_{12})$$

$$1 + z + z^2 + \dots + z^{11} = (1 - z_1)(1 - z_3)(1 - z_4) \dots (1 - z_{12})$$

$$\text{Put } z = 1, 12 = |1 - z_2| |1 - z_3| |1 - z_4| \dots |1 - z_{12}|$$

$$\Rightarrow |z_2 - z_1|^2 = |z_1|^2 |(e^{i2\pi/12} - 1)|^2 = 2^2 \sin^2 \frac{\pi}{12} \Rightarrow |A_1 A_2|^2 + |A_1 A_3|^2 + \dots + |A_1 A_{12}|^2$$

$$= 4 \left[\sin^2 \frac{\pi}{12} + \sin^2 \frac{2\pi}{12} + \sin^2 \frac{3\pi}{12} + \dots + \sin^2 \frac{11\pi}{12} \right] = 2 \left[11 - \left(\cos \frac{2\pi}{12} + \cos \frac{4\pi}{12} + \dots + \cos^2 \frac{2\pi}{12} \right) \right]$$

$$= 2(11 - (-1)) = 24$$

56. 04.00

$$\text{Sol. } \frac{4}{7} \times \frac{{}^8C_5}{{}^{12}C_5}$$

57. 09.00

$$\text{Sol. } \frac{4}{12} \times \frac{3}{11} = \frac{1}{11}$$